

Movie RAG: End-to-End Generation Evaluation

Combined Movies Run

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1 Evaluation scope

This report evaluates the final grounded answer output rather than treating retrieval coverage as an answer-correctness proxy. All 40 questions were forced through the B3 semantic path: production hybrid retrieval, weak-query escalation, evidence gating, and grounded OpenAI generation. Generated answers were graded against the golden answer and evidence using a strict structured judge. Citation validity additionally required syntactically valid context numbers.

The generation and judge model was `gpt-4.1-mini`. The golden CSV is still AI-drafted with `human_verified=false`; therefore these are exploratory results, not official assessment scores.

2 Main results

Config	R@5	R@20	MRR	nDCG@10	Ans.	Refuse	Over	p95 ms
B3 retrieval proxy	1.000	1.000	1.000	1.000	1.000	0.091	0.000	1532.1
B3 final + LLM	1.000	1.000	1.000	1.000	0.862	0.909	0.138	4093.7

There were 29 answerable and 11 non-answerable items. The final system answered 25 of 29 answerable questions correctly, producing answer correctness of 0.862. It correctly refused 10 of 11 non-answerable questions, producing a correct-refusal rate of 0.909. It incorrectly refused 4 of 29 answerable questions, producing an over-refusal rate of 0.138.

Citation validity was 0.793: 23 of 29 answerable items had a fully valid final citation result. The two non-refusal citation errors were correct answers with an additional citation to a context item that did not support the stated fact. The remaining four citation failures were the four over-refusals.

3 Development and test splits

Split	Items	Ans.	Citation	Refuse	Over	p95 ms
Development	24	0.824	0.765	0.857	0.176	4361.9
Test	16	0.917	0.833	1.000	0.083	3598.9
All	40	0.862	0.793	0.909	0.138	4093.7

4 Observed failures

ID	Label	Observation
G004	Over-refusal	Confident retrieval found The Orchard Below, but generation refused the director question.
G005	Over-refusal	Confident retrieval found The Orchard Below, but generation refused the writer question.

G007	Over-refusal	Confident retrieval found Ember Transit, but generation refused the director question.
G020	Over-refusal	Retrieval found both required movies, but generation refused the combined-runtime calculation.
G006	Invalid citation	Runtime was correct, but context item 2 did not support the claim.
G014	Invalid citation	Genres were correct, but context item 2 did not support the claim.
G038	Missed refusal	“Who directed it?” had no conversation context, but the model selected an unrelated movie instead of requesting clarification.

These results demonstrate that perfect retrieval does not imply perfect final answers. The principal final-stage problems are generator over-refusal, extraneous citations, and failure to reject an unresolved conversational reference.

5 Latency and API usage

Answer latency includes retrieval, escalation, and final generation, but excludes the evaluation judge call. Median answer latency was 2432.3 ms and p95 was 4093.7 ms.

The complete run, including answer grading, made 65 OpenAI calls and used 121,309 input tokens plus 3,828 output tokens, for 125,137 total tokens. USD cost is not reported because model-specific pricing was not configured in the evaluator.

6 Uncertainty and limitations

The answer-correctness estimate was 0.862 with a 95% paired-bootstrap interval of [0.724, 0.966]. Correct refusal was 0.909 with interval [0.727, 1.000], using 5,000 resamples.

- The same model generated and graded answers, which can introduce self-judging bias.
- The golden set has not been human-verified.
- Questions were forced through semantic generation. Normal production auto-routing would answer several factual entity questions directly from structured records without an LLM.
- Retrieval metrics are document-ID metrics; citation and answer scores evaluate the final generated output.
- The test split was used exploratorily and is not the rubric’s frozen, once-only official test run.

7 Artifacts

The full generated answers, retrieval IDs, judge rationales, citations, latencies, escalation decisions, and token counts are stored in `per_item_generation.csv`. Aggregate values are stored in `metrics_generation.csv` and `metrics_generation.json`. The seven observed failures are summarized in `failure_analysis_generation.csv`.